

DiagAccU

Interface Documentation

Version 2.0.1

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1. Introduction

This guide describes the inputs and controls available in the distributed, compiled DiagAccU 2.0.1 interface. Available inputs depend on the selected plot or table.

The test is positive for values at or above the diagnostic threshold t . Population-parameter sampling, prevalence, and measurement uncertainty sources are assumed independent, with covariance retained between the estimated mean and standard deviation within each population. Their propagated variance contributions are added, and the combined standard uncertainty is the square root of the resulting variance. Joint propagation of measurement uncertainty retains the covariance between sensitivity and specificity. The constant coefficient b_0 , proportional coefficient b_1 , and selected model specify the measurement standard uncertainty.

2. Interface Controls

2.1. Tabbed Navigation

Choose a primary tab and then the required submodule. Each terminal view has its own controls and saved defaults; changing a control in one view does not synchronize every other view.

2.2. Numerical Settings: Sliders

The program offers controls for numerical settings, which can be adjusted through sliders. Press the 'alt' or 'opt' key while dragging the mouse for more precise control. For even finer adjustments, also press the 'shift' or 'ctrl' keys.

2.3. Non-Numerical Settings

Buttons and drop-down menus select the measure, distribution family, measurement-uncertainty model, and other settings. Available controls depend on the selected view.

2.4. Display Controls

2.4.1. Plots

Extended and limited settings principally change the range of the horizontal axis. Threshold plots span the population means plus or minus three or one standard deviations, respectively, with a nonnegative rounded lower bound. Prevalence profiles

use 0.001–0.999 or 0.01–0.2. Profiles of b_0 use 0–0.5 times s_H or 0–0.15 times s_H ; profiles of b_1 use 0–0.1 or 0–0.05. The plotted sample-size range is generated automatically. These settings do not specify the range over which the measure is optimized.

2.4.2. Tables

Table views provide a decimal-display control from 1 to 10, with a default of 3. This changes displayed precision, not the precision of the underlying calculation.

3. Input Variables and Parameters

The following ranges describe interface controls. Diseased and nondiseased populations are denoted D and H, and prevalence is v . Arithmetic means and Bessel-corrected standard deviations are supplied for all population families.

t : $\max(0.01, \min(m_H - 6s_H, m_D - 6s_D)) - \max(m_H + 6s_H, m_D + 6s_D)$

n_D : 2–10,000

m_D : 0.1–10,000

s_D : 0.01–1,000

n_H : 2–10,000

m_H : 0.1–10,000

s_H : 0.01–1,000

v : 0.001–0.999 in the views that expose prevalence; other views derive $v = \frac{n_D}{n_D + n_H}$ from the sample counts.

n_U : 20–10,000; interval views use this control to assign $v_{\text{meas}} = n_U - 1$ to the common measurement-uncertainty source.

b_0 : $0 \leq b_0 \leq s_H$

b_1 : 0–0.1

$1 - \alpha$: 0.900–0.999

n : 20–20,000; in the sample-size profile, the total sample size is varied automatically while prevalence is held fixed at the selected value.

The diagnostic threshold, both means, both standard deviations, and b_0 must use the same measurement unit; b_1 is dimensionless. The measurement standard uncertainty uses that same unit.

In uncertainty profiles parameterized by total sample size and prevalence, the program uses $n_D = vn$, $n_H = (1 - v)n$. Both resulting stratum sizes must exceed one for the core calculation to be valid. Thus some extreme prevalence–sample-size combinations within the slider ranges can produce unavailable results. Noninteger effective sample sizes can occur in calculations over the plotted range.

5. Notation

5.1. Variables and Parameters

t : diagnostic threshold

n : total sample size

v : prevalence

b_0 : constant contribution to the common measurement standard uncertainty

b_1 : proportional coefficient for the common measurement standard uncertainty

$1 - \alpha$: confidence level

5.2. Diagnostic Accuracy Measures

Se: sensitivity

Sp: specificity

ACC: overall diagnostic accuracy

PPV: positive predictive value

NPV: negative predictive value

DOR: diagnostic odds ratio

LR⁺: positive likelihood ratio

LR⁻: negative likelihood ratio

J: Youden's index

BA: balanced accuracy

CZ: concordance probability

FMI: Fowlkes–Mallows index

κ : Cohen's kappa

PABAK: prevalence- and bias-adjusted kappa.

F_1 : F_1 score.

MCC: Matthews correlation coefficient

5.3. Uncertainty and Interval Outputs

u_c : combined standard uncertainty

u_s : total sampling standard uncertainty

u_m : measurement standard uncertainty of the selected measure obtained by propagation

LCL: lower confidence limit

UCL: upper confidence limit

QC: quality control

6. Citation:

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